Class Manual 4

1. **Count students per course (sorted by popularity)**

   QUERY

```
SELECT C.course_name, COUNT(E.student_id) AS total_students
FROM COURSE C
LEFT JOIN ENROLLMENT E ON C.course_id = E.course_id
GROUP BY C.course_name
ORDER BY total_students DESC;
```

   OUTPUT

course_name	total_students
DBMS	6
OS	3
Networks	2
Data Science	2
AI	1

2. **Find courses with low enrollment (including zero)**

   QUERY

```
SELECT C.course_name, COUNT(E.student_id) AS total_students
FROM COURSE C
LEFT JOIN ENROLLMENT E ON C.course_id = E.course_id
GROUP BY C.course_name
HAVING COUNT(E.student_id) <= 1
ORDER BY total_students ASC;
```

   OUTPUT

course_name	total_students
AI	1

3. **Find average CGPA per course for enrolled students**

   QUERY

```
SELECT C.course_name, ROUND(AVG(S.cgpa), 2) AS avg_cgpa
FROM ENROLLMENT E
JOIN STUDENT S ON E.student_id = S.student_id
```

```
JOIN COURSE C ON E.course_id = C.course_id
GROUP BY C.course_name;
```

OUTPUT

course_name	avg_cgpa
DBMS	8.43
OS	8.47
Networks	7.55
AI	7.50
Data Science	9.40

4. **Display grade distribution per student (highest CGPA first)**

QUERY

```
SELECT S.name, S.cgpa, C.course_name, E.grade
FROM STUDENT S
JOIN ENROLLMENT E ON S.student_id = E.student_id
JOIN COURSE C ON E.course_id = C.course_id
ORDER BY S.cgpa DESC;
```

OUTPUT

name	cgpa	course_name	grade
Neha	9.50	DBMS	A
Neha	9.50	Data Science	A
Priya	9.30	Data Science	A
Riya	9.10	DBMS	A
Riya	9.10	OS	A
Aman	8.50	DBMS	A
Aman	8.50	OS	B
Sneha	8.20	DBMS	B
Sneha	8.20	Networks	B
Karan	7.80	DBMS	B
Karan	7.80	OS	B
Arjun	7.50	DBMS	C
Arjun	7.50	AI	A
Rahul	6.90	Networks	C

5. **Find out high CGPA courses with enrollment > 1**

QUERY

```
SELECT C.course_name,
       ROUND(AVG(S.cgpa), 2) AS avg_cgpa,
       COUNT(*) AS total_students
FROM ENROLLMENT E
JOIN STUDENT S ON E.student_id = S.student_id
JOIN COURSE C ON E.course_id = C.course_id
GROUP BY C.course_name
HAVING COUNT(*) > 1
ORDER BY avg_cgpa DESC;
```

OUTPUT

course_name	avg_cgpa	total_students
Data Science	9.40	2
OS	8.47	3
DBMS	8.43	6
Networks	7.55	2